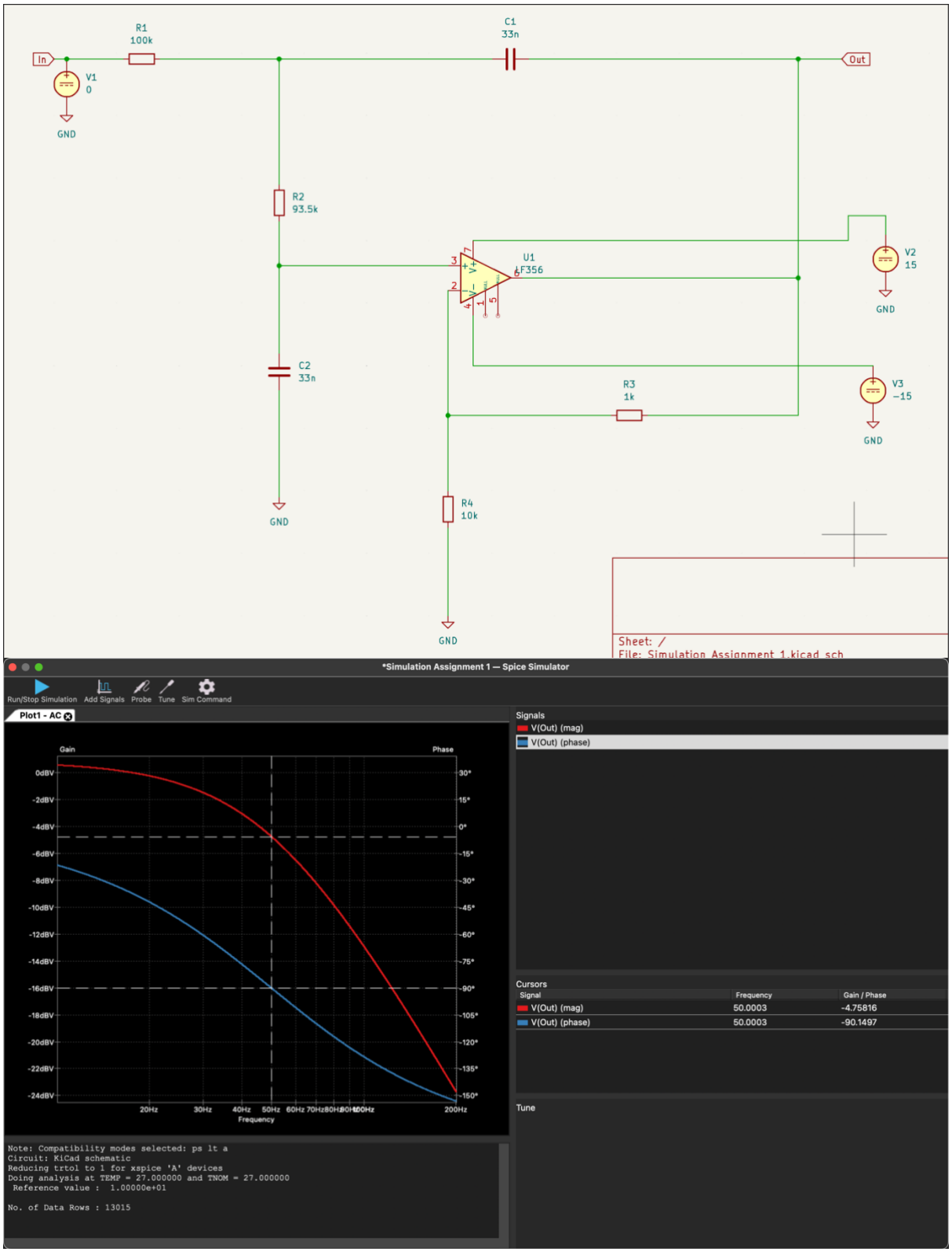
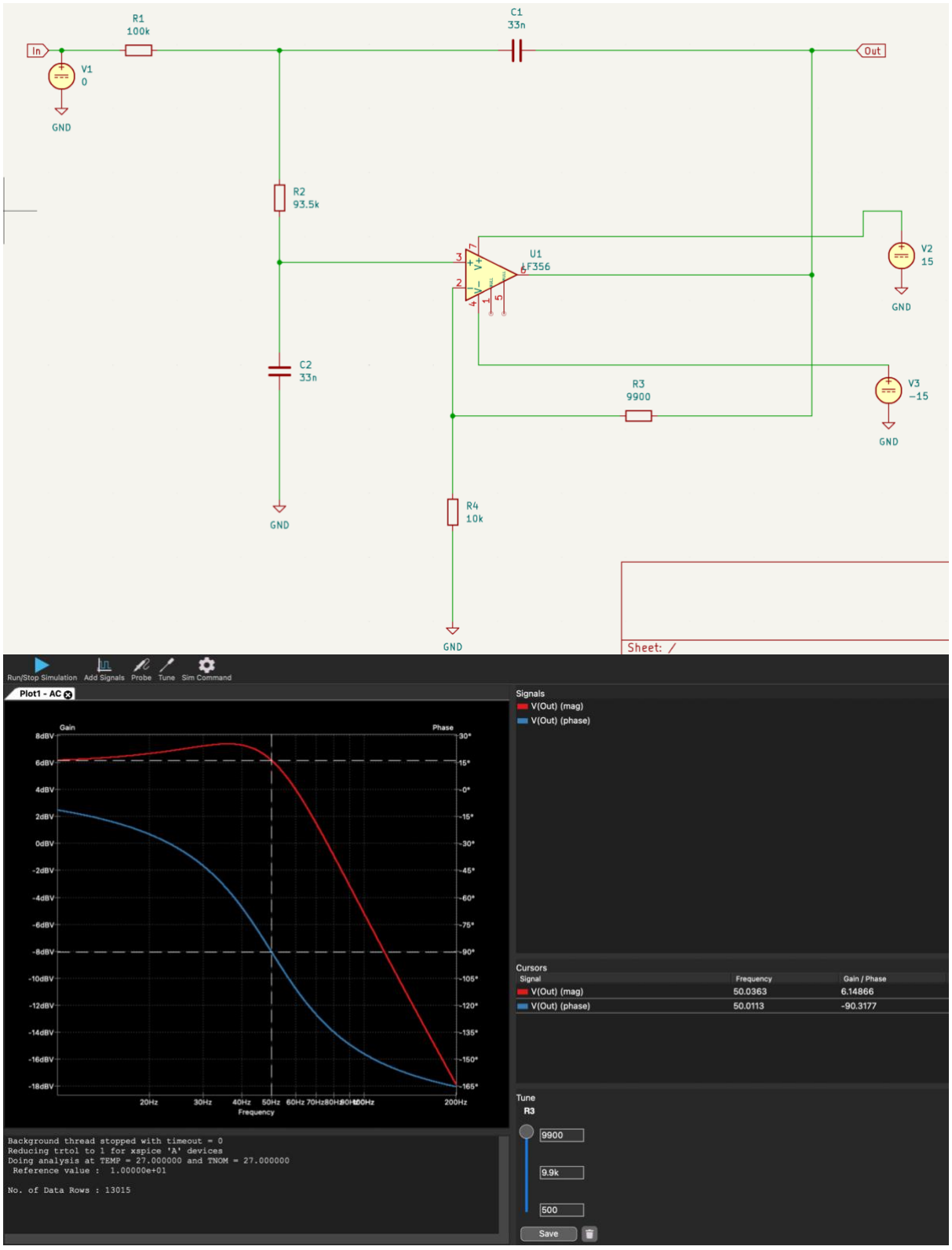
Simulation Assignment 1

PART A:



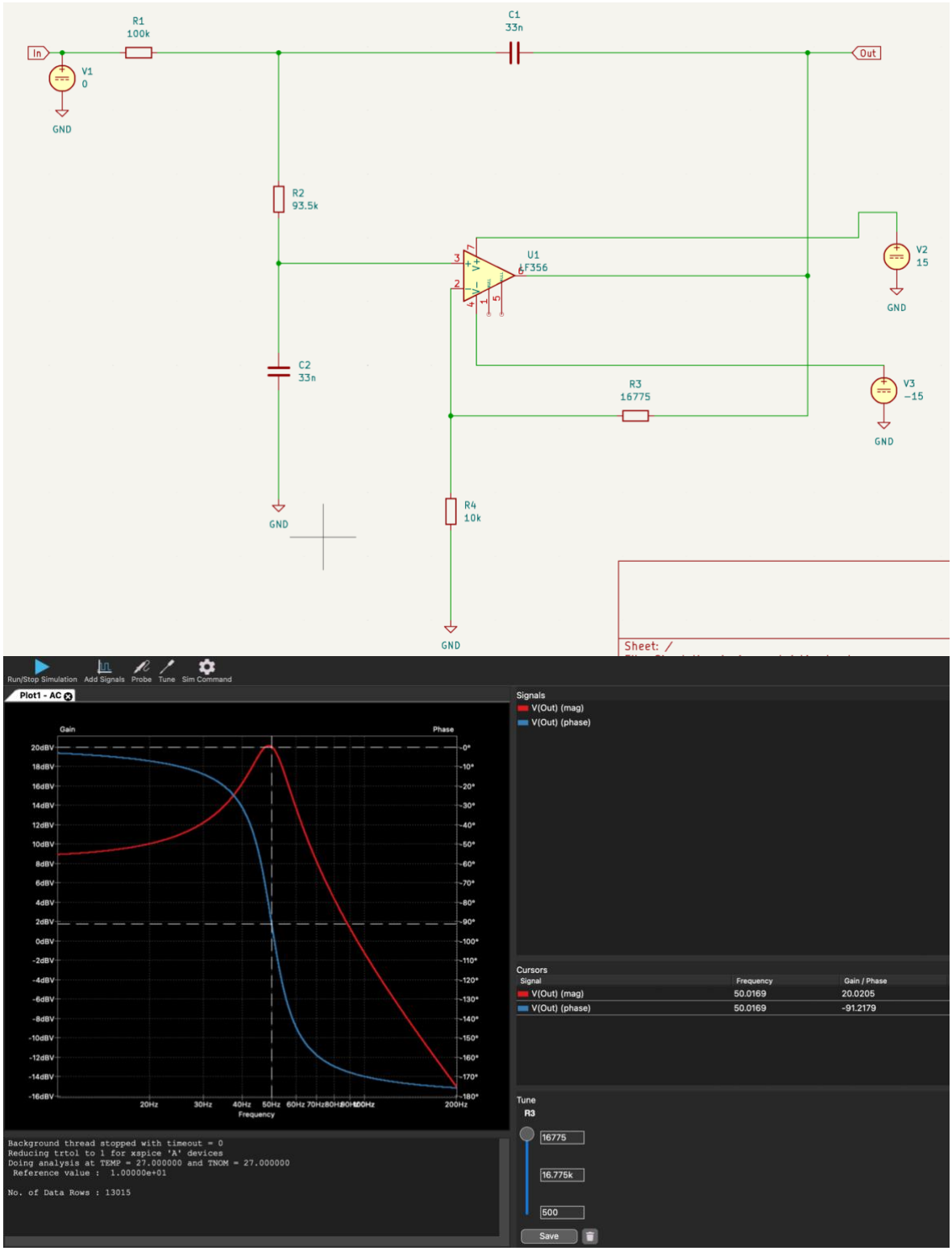
- Value of gain magnitude [dB] at 50 Hz = -4.75816 dB
- Value of phase [deg] at 50 Hz = -90.1497

PART B:



- Value of gain magnitude [dB] at 50 Hz = 6.14866 dB
- Value of phase [deg] at 50 Hz = -90.3177
- When R_3 is tuned to 9900 ohms, we will have frequencies around 50 Hz to be passed more than other frequencies.

PART C:



• Value of gain magnitude [dB] at 50 Hz = 20.0205 dB
• Value of phase [deg] at 50 Hz = -91.2179
• The slope becomes steeper at 50 Hz due to resonance response. Due to low impedance the phase curve will change rapidly at resonant frequency.
• The 3 dB bandwidth here is 44-53 Hz ($\Delta = 9$ Hz)